

Valencia Sound Craft

Portfolio site + interactive rhythm and harmony tool system

WHAT IT IS

Valencia Sound Craft is a React/Vite web app that combines a musician/producer portfolio site with direct-linkable rhythm and harmony tool routes.

Repo copy frames it as a creative music technology site for practice, composition, and cross-cultural rhythm exploration.

WHO IT'S FOR

Primary user/persona: musicians, guitarists, bassists, students, and creative technologists who want to learn, practice, compose, or explore rhythm and harmony interactively.

WHAT IT DOES

- Homepage with portfolio, services, about, contact, and tool entry points.
- Direct routes for Rhythm Engine, Harmony Lab, Rhythm Map, Circle of Fifths, and Tonnetz.
- Shared musical state keeps key, mode, tempo, rhythm, region, chord progression, and transport aligned.
- Rhythm atlas with country/region browsing, sequencer playback, tempo control, and quick rhythm selection.
- Harmony workspace with mode visualization, fretboard, keyboard, sheet music, metronome, and progression tools.
- Real-time playback utilities plus pattern morphing and MIDI export in the rhythm engine/drum machine.
- SEO-ready tool pages with canonical tags and SoftwareApplication structured data.

HOW IT WORKS

- App shell: ``src/main.tsx`` mounts a React SPA; ``src/App.tsx`` wires React Router, React Query, toasts, and ``GlobalMusicProvider``.
- Page composition: ``/`` renders marketing/portfolio sections; ``/tools/*`` lazy-loads five workspaces through ``ToolsLayout``.
- State layer: ``src/state/globalMusicState.tsx`` stores key, mode, tempo, rhythmId, region, chord progression, and single-owner transport state.
- Tool adapters: ``useRhythmSelection`` and ``useHarmonySelection`` bridge route UIs to the shared store; route ``useTool`` hooks feed each workspace.
- Engines/data: Rhythm and Map use ``GlobalRhythmEngine``, local rhythm catalogs, and a React-Leaflet country map; Harmony, Circle, and Tonnetz use ``ModeVisualizer`` and theory components.
- Backend/services: React Query is initialized. A generated Supabase client and SQL seed scripts exist; active frontend Supabase usage: Not found in repo.

HOW TO RUN

- Install dependencies: ``npm install``
- Start the app: ``npm run dev``
- Open the local Vite URL shown in the terminal.
- Optional checks: ``npm run build`` and ``npm test``
- Node version pin and env setup instructions: Not found in repo.